

Tabletop Course Marshal

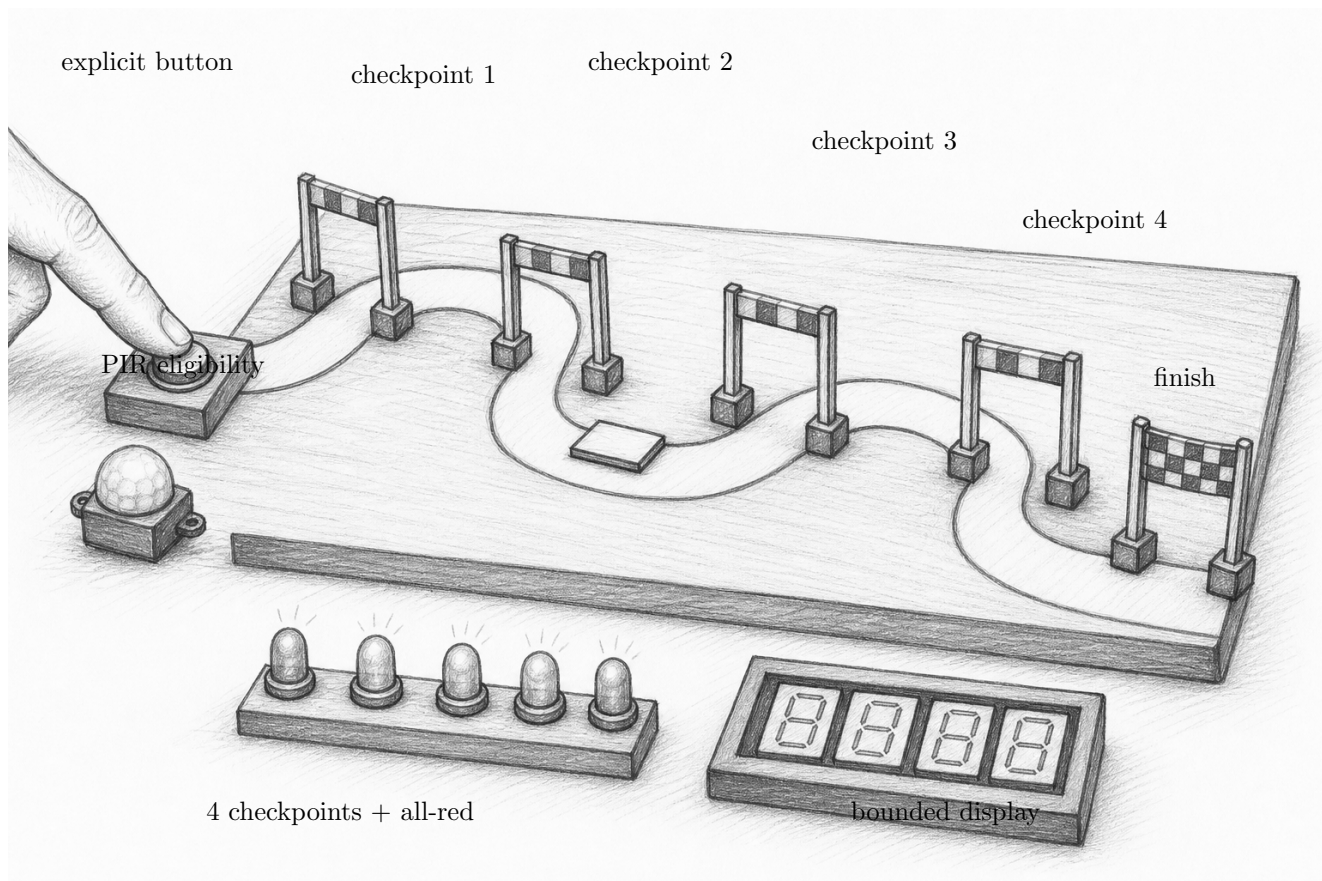
Authorize once, cross in order, freeze the evidence

E0 host verified; powered fixture and bench acceptance open

Lesson	042 — project	Time	150–210 minutes
Prerequisites	Lessons 040–041	Board	deterministic Mega replay
Evidence	checkpoint LEDs, heartbeat, all-red, display	Boundary	hand-moved card; no actuator

Authority rule. PIR motion can make a start eligible, but it can never start a run. Only a debounced press from the configured button, in the same course-clock frame as fresh valid PIR Motion evidence, emits the start event. A press outside that frame is not remembered.

1 A course made from semantic evidence



Alternative text: a pencil drawing shows a finger pressing the separate start button, a nearby PIR eligibility witness, a hand-moved card crossing four ordered gates and a finish, then checkpoint LEDs, an all-red indicator, and a display.

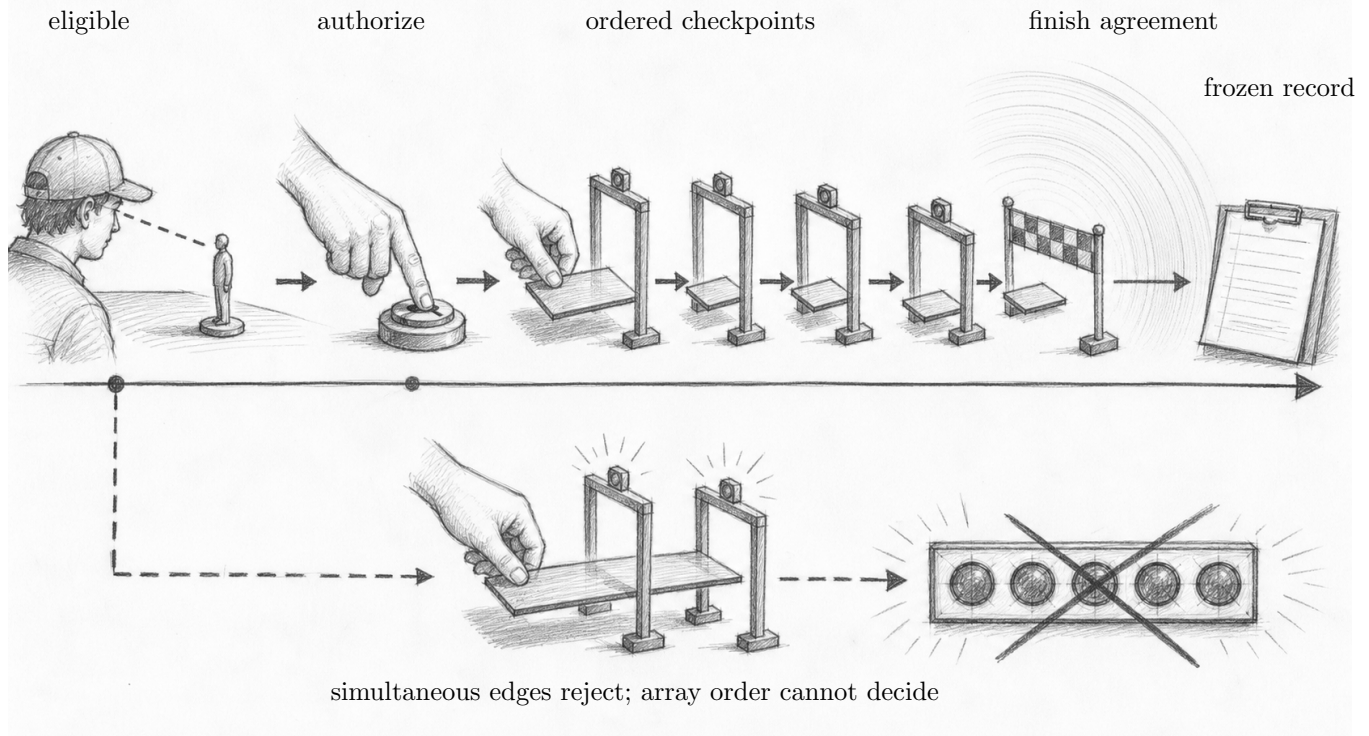
The policy supports four checkpoints; the canonical replay configures three.

The project consumes qualified copied values rather than pins or retail modules. Configuration maps one through four unique checkpoint IDs to optical source kind, source ID, and calibration revision. That semantic binding, not input-array or pin order, defines the course.

`CourseStartPolicy` receives the copied button edge and PIR state in one frame. `CourseMarshal` receives its start event, one immutable `PresenceSnapshot`, and at most four qualified checkpoint events. It validates

the entire frame before mutation. Changed same-time, backward, and half-range-invalid frames fault without accepting a partial checkpoint.

2 Walk the evidence timeline



Alternative text: a pencil timeline separates PIR eligibility from the required button press, continues through the configured checkpoints and finish agreement to a record, and branches to all-red when gates collide. The policy supports four checkpoints; the canonical replay configures three.

Start in **Disarmed**. A structurally valid frame enters **Arming**; fresh valid PIR Motion evidence makes the marshal **Ready**. PIR alone leaves it there. Only a same-frame explicit start event enters **Running**. A press before or after eligibility does nothing and is never latched.

Each accepted checkpoint advances one semantic slot. Two distinct legal events whose timestamps are inside the inclusive simultaneity window reject as ambiguous. A repeated identity is idempotent, while a distinct event for an already accepted checkpoint rejects as duplicate.

Finish acceptance requires the complete checkpoint prefix, a fresh valid finish-guard activation event, and fresh valid approach-range evidence within the agreement window. The resulting terminal record stays immutable until `acknowledgeRecord()` or reconstruction.

3 Fixed storage is an architectural choice

The original monolithic design measured 653 bytes for the marshal and 432 bytes for its presenter on AVR, exceeding the 128-byte largest-object ceiling. The remediated design uses six explicit caller-owned fixed storages:

Budget item	AVR bytes
run, trigger, trigger-presence, and three replay storages	456
six storages plus marshal	525
storages plus marshal, start policy, and presenter	639

Every individual public type and policy object now measures at most 128 bytes. The marshal retains references to all six storages for its full lifetime; input-view pointers are borrowed only for `update()`. There is no heap, RTC, SD card, or persistence claim. Reconstruction begins a new volatile identity epoch. Sequence exhaustion is permanent for one object lifetime.

These are E0 policy subtotals, not a complete Mega budget. The canonical replay sketch measures 15,596 bytes of flash and 901 bytes of static SRAM on the Mega 2560. Powered composition still requires aggregate current, pin, cadence, and stack evidence.

4 Observe without Serial

Visible path	Prediction and interpretation
three checkpoint LEDs	canonical replay evidence; policy supports four
ready/heartbeat	eligible readiness and supplied-time progress
all-red	rejected or faulted run; inspect the retained trigger
four-digit display	one bounded cell per update; elapsed or fault code

`CourseMarshalPresenter` copies only the compact marshal snapshot. It advances at most one display cell per later timestamp, so a sparse update cannot monopolize the loop. Display or renderer failure cannot alter course phase, order, elapsed time, or the frozen record.

5 Predict, observe, interpret

1. Supply valid PIR Motion evidence without a button press. Predict **Ready**, not **Running**.
2. Press the configured button in the same eligible frame. Predict exactly one start event. Repeat a held input and predict no second event.
3. Cross checkpoint IDs in configured order. Predict one additional local LED after each accepted event.
4. Present two distinct checkpoint events within the simultaneity window. Permute their array positions; predict the same rejection and all-red.
5. Reach the finish guard early or without fresh approach evidence. Predict rejection with trigger evidence preserved.
6. Complete the ordered course and agreeing finish. Predict a stable accepted record and elapsed display until acknowledgment.
7. Inject failed source status or malformed time. Predict a fault without partially advancing the accepted prefix.

6 Powered boundary and shutdown

Lesson 042 publishes no wiring table and no electrically authoritative schematic. Exact optical, PIR, HC-SR04, button, display, and indicator specimens must first close identity, pin order, voltage, output-rail, polarity, pull-up, emitter, current, resource, timing, stack, and aggregate SRAM gates. Only then may a separately reviewed E1 composition add pin-by-pin prose and a formal schematic.

The physical model is moved only by hand. There is no motor, powered gate, vehicle control, launcher control, or stored-energy command. Disconnect USB before wiring changes and stop for heat, odor, reset, unstable rail, or unexpected output.

Resource acquisition and electrical safe state are separate evidence. A future qualified fixture must prove claims and rollback independently from inactive indicators and high-impedance sensor endpoints after shutdown.

Physical acceptance remains open. Compilation, replay, object-size measurements, pencil drawings, and documentation do not prove a sensor, current budget, display, timing cadence, stack peak, or safe shutdown on a bench. Leave the acceptance card blank until those observations exist.

7 Software evidence

Host tests cover authorization timing, eligibility phases, checkpoint order, simultaneity, replay identity, finish agreement, timeout, sequence exhaustion, record acknowledgment, presenter pacing, rollover, reset, and source faults. The canonical Mega sketch is a deterministic copied-evidence replay; it is not a claim that any powered module is qualified.

- ADK Lessons 040–042 optical course marshal design record.
- ADK deterministic testing contract and safety model.
- ADK component architecture stress-pass template.

This untagged print edition has a searchable HTML alternative. It makes no PDF/UA or physical accessibility claim.